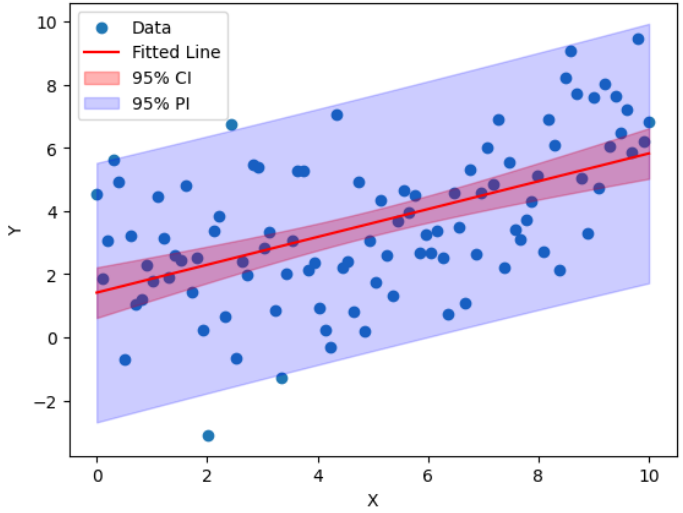
Lecture 23

Uncertainty Quantification I



Announcements

- Course project feedback out tomorrow
  - Directed questions out later this week

- Office hours every Tuesday (noon to 4pm) and Friday (11am to 1pm)
  - Link to book appointments is on the course website

Last Week

Statistical Inference

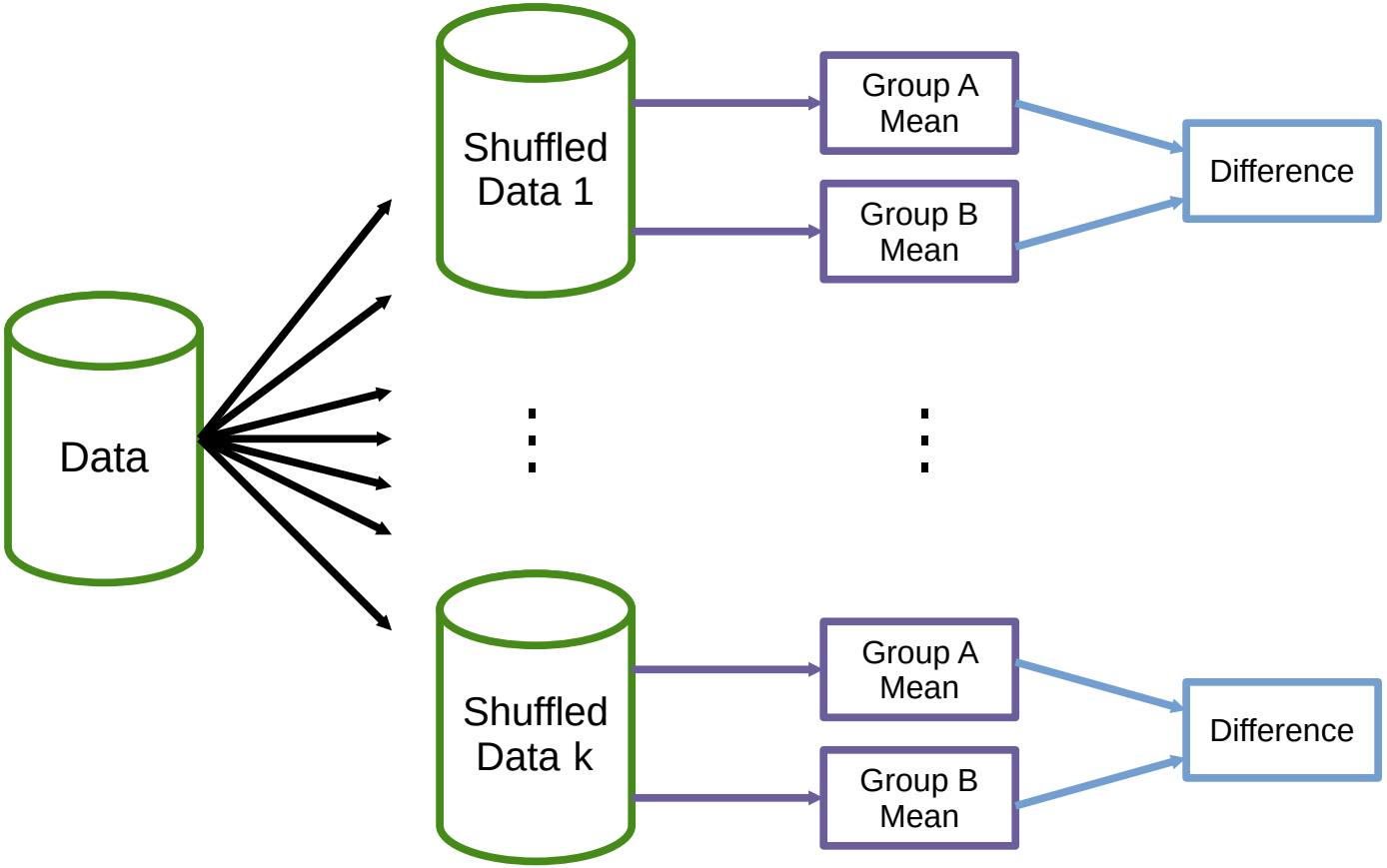
- Sampling
- Sample Variance
- Hypothesis Testing (via permutation tests)

Today

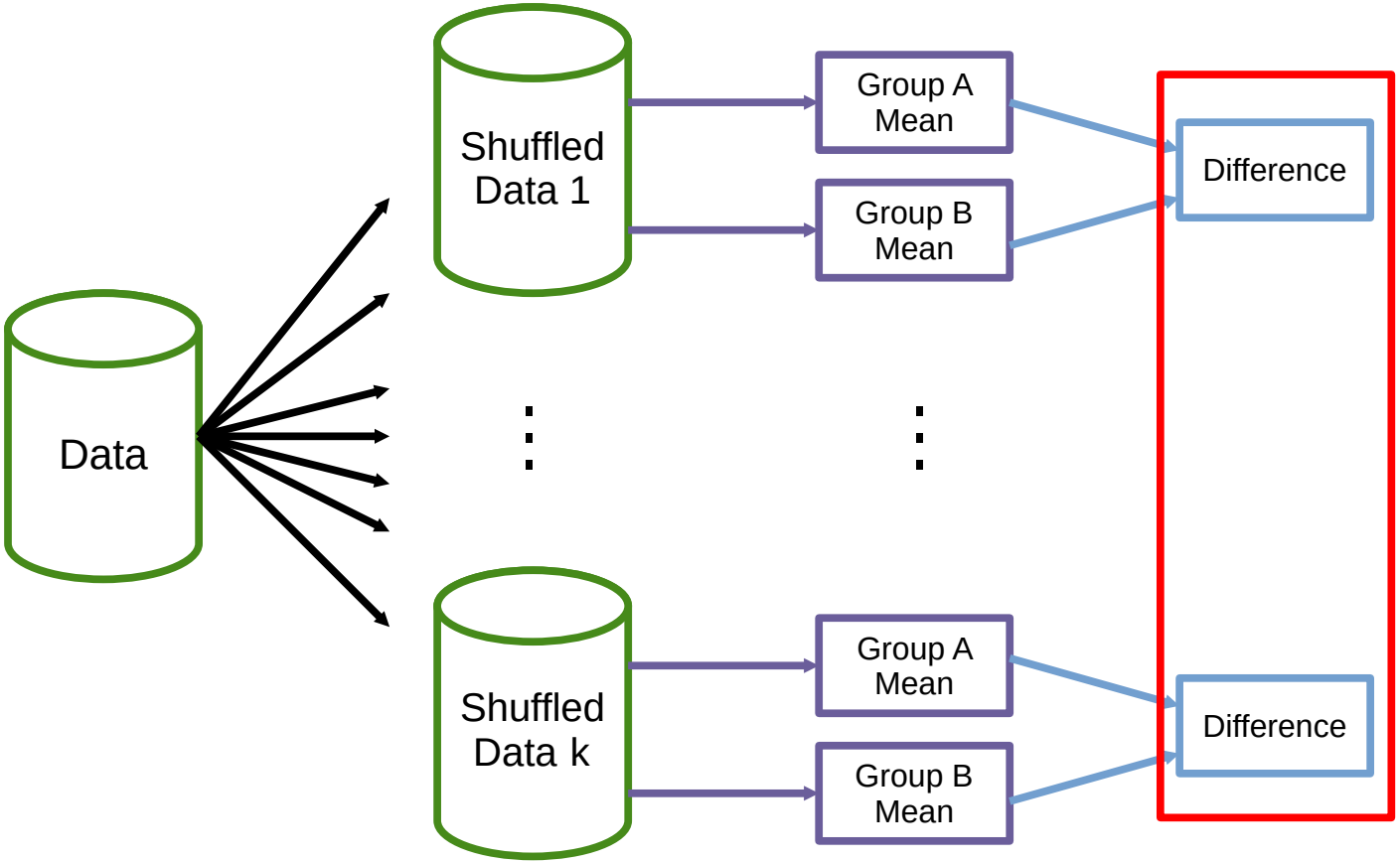
Confidence Intervals

- Idea
- Bootstrapping

Permutation Test



Permutation Test

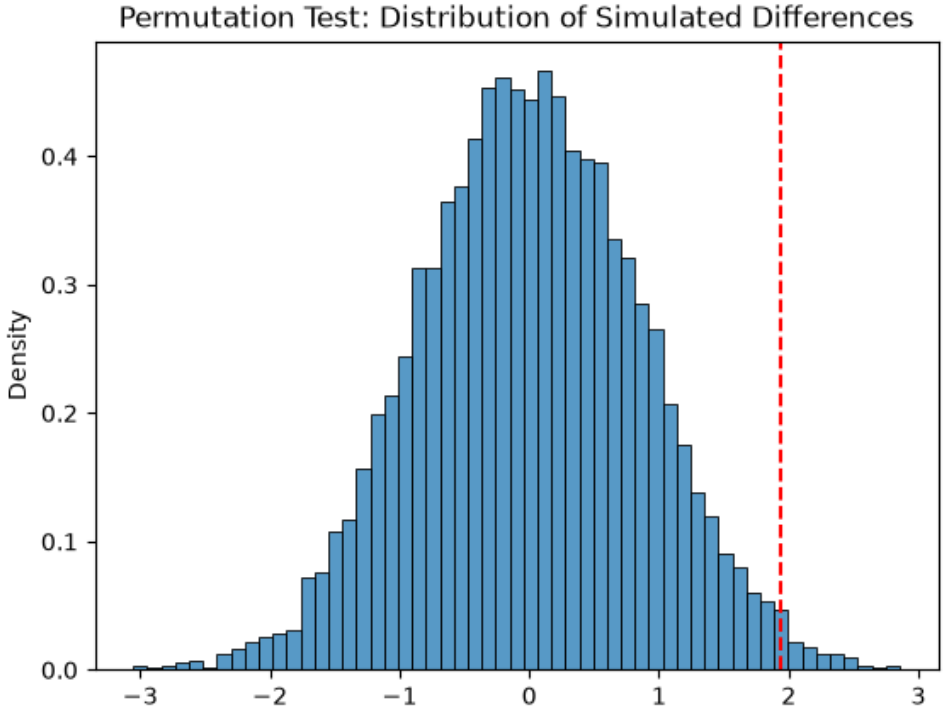


Analyze distribution of differences

Mobile vs Walk-in Wait Times

Average wait times from our sample

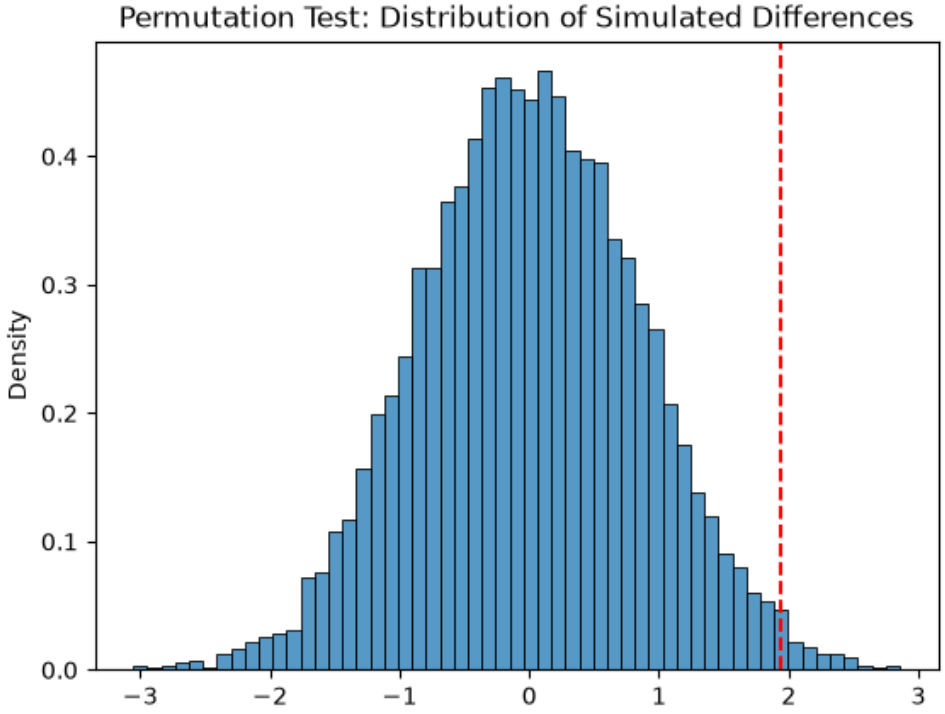
- Mobile: 3.87
- Walk-in: 5.80
- Difference: 1.93



Mobile vs Walk-in Wait Times

Only 1.14% of shuffled datasets produced a gap as big as the observed gap

- If there were no differences between subgroups, the probability that an effect at least as large as the observed effect is approx. 0.0114
- We typically set the cut off for saying the effect is real at 0.05
- A version of a "p-value"
- In this case, we would **reject** the null hypothesis



Probability

What does probability mean?

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One interpretation

- Relative frequency of event occurring (frequentist interpretation)

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  - a possible outcome: coin flip landing heads, mean > 1

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$$\frac{\#\text{ times event occurs}}{\#\text{ trials total}}$$

Probability

What does probability mean?

One interpretation

- Relative frequency of event occurring (frequentist interpretation)
  - a possible outcome: coin flip landing heads, mean > 1

$$\frac{\text{\# times event occurs}}{\text{\# trials total}} \qquad \frac{\text{\# heads}}{\text{\# coin flips}} = \text{P(heads)}$$

Probability

What does probability mean?

One interpretation

- Relative frequency of event occurring (frequentist interpretation)
  - a possible outcome: coin flip landing heads, mean > 1

$$\frac{\text{\# times event occurs}}{\text{\# trials total}} \qquad \frac{\text{\# times mean > 1}}{\text{\# total samples}} = P(\text{mean} > 1)$$

Probability

Long-run Guarantee

- Describes a process rather than a single sample

Law of Large Numbers

- Statistics from small samples are susceptible to noise (from sampling variance)
- In the long run, this noise cancels out

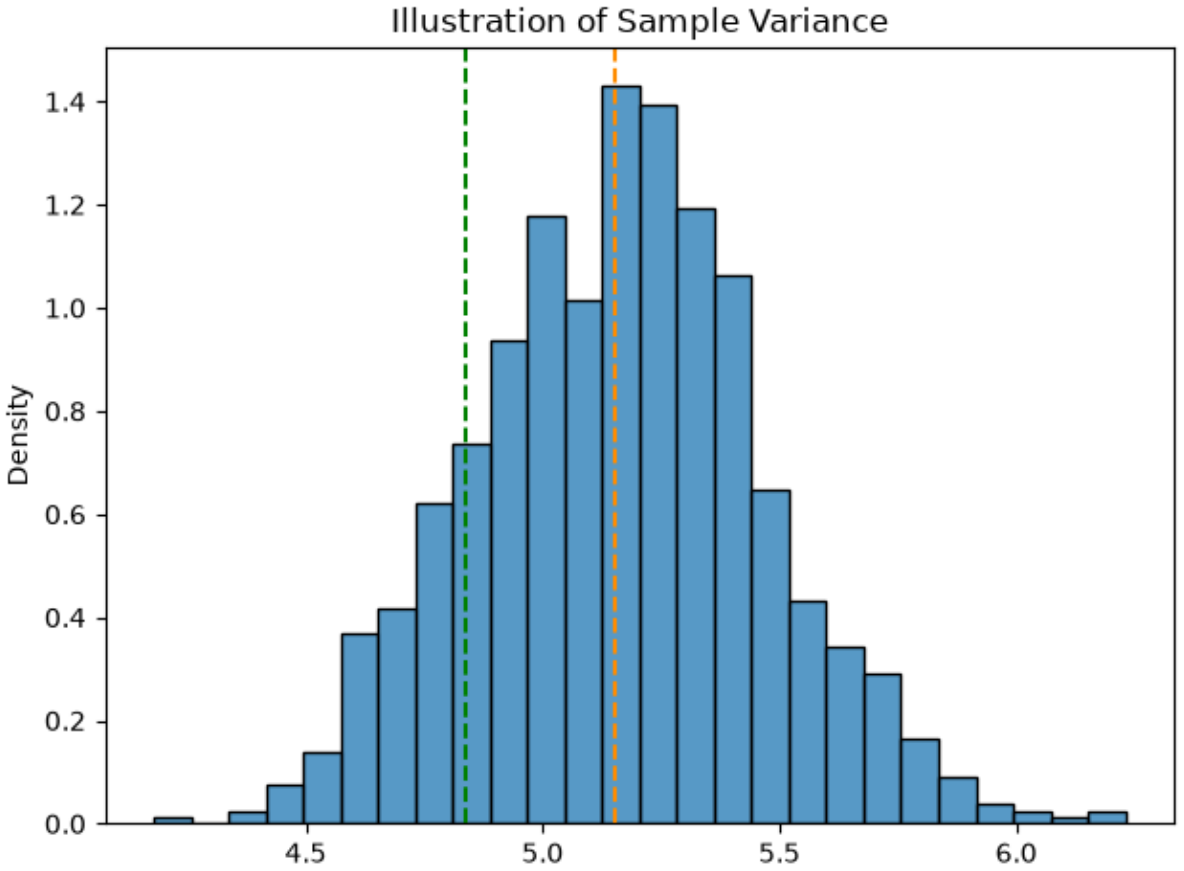
Starbucks Dataset

Average wait time

- Population: 5.15
- Sample: 4.82

Distribution of Sample Means

- Varies between about 4.5 and 6
- Mean of sampling distribution (5.16) approximates population mean (sampling is unbiased)



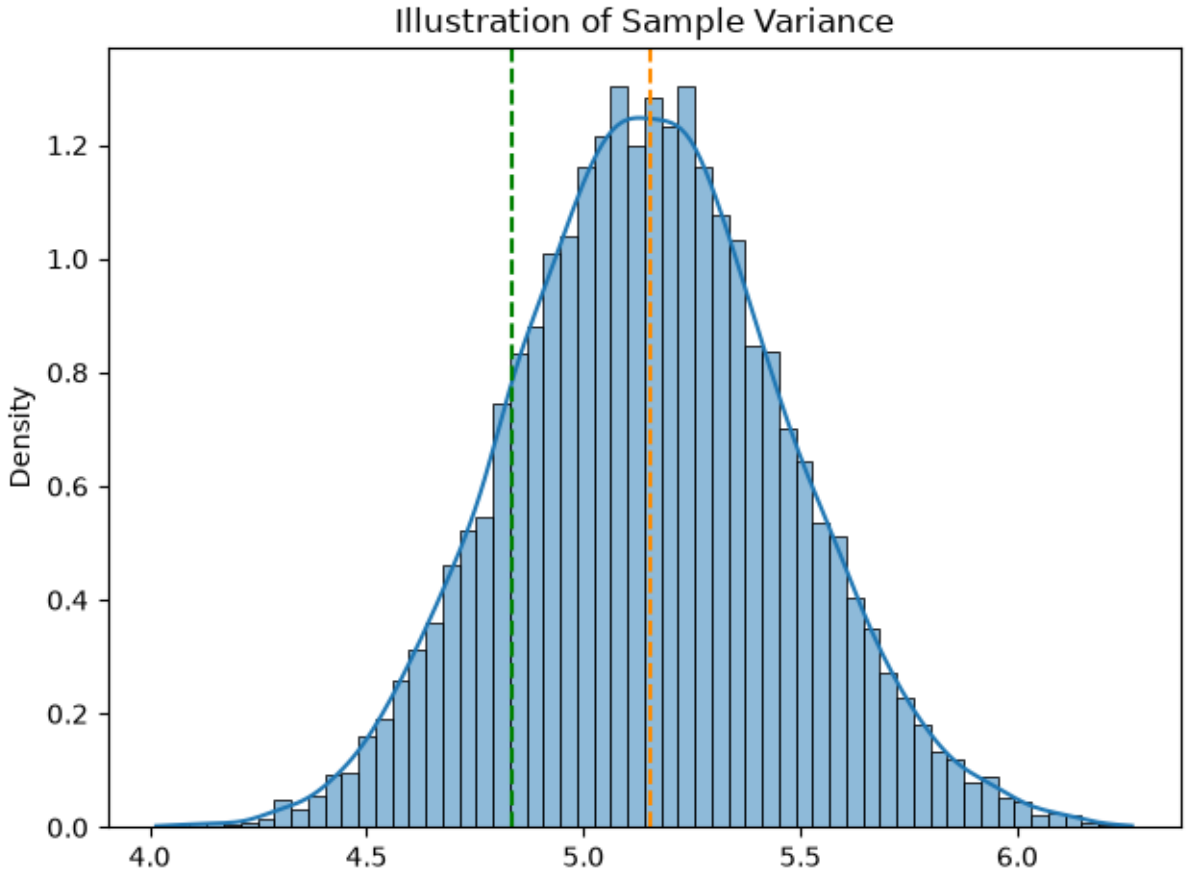
Starbucks Dataset

Average wait time

- Population: 5.15
- Sample: 4.82

Distribution of Sample Means

- Varies between about 4.5 and 6
- Mean of sampling distribution (5.16) approximates population mean (sampling is unbiased)
- Central limit theorem

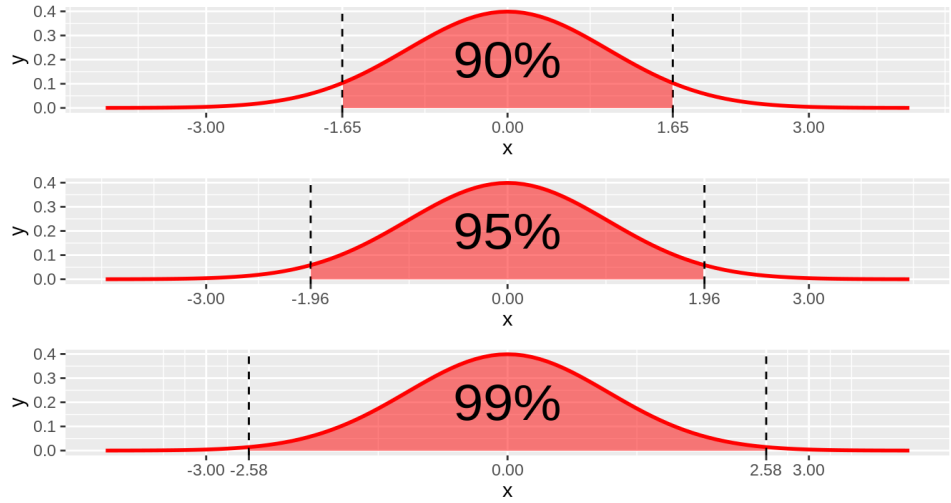


Confidence Interval

Probabilistic lower and upper bounds on a statistic

For example:

- `(1, 3)` a 95% confidence interval for k
- We think it is likely that the population statistic k is between 1 and 3
- Typically, the higher our confidence, the wider our interval gets

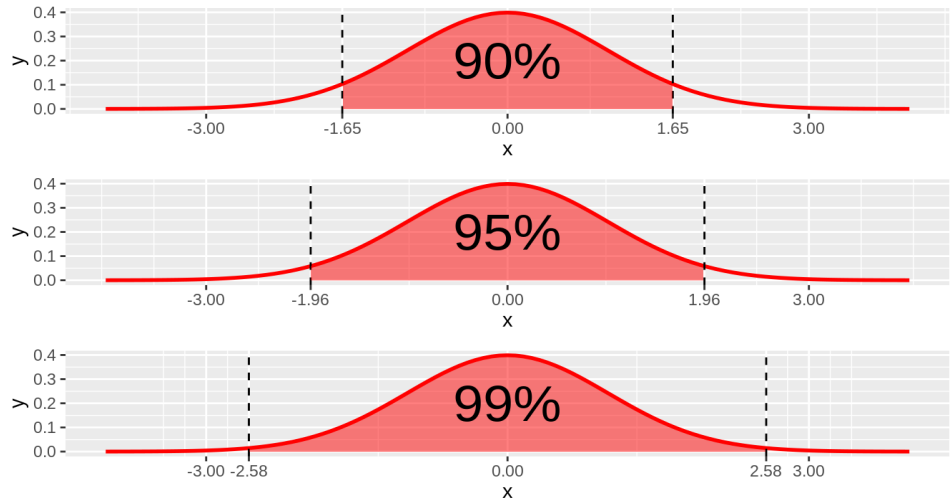


Confidence Interval

Probabilistic lower and upper bounds on a statistic

Interpretation:

- `(1, 3)` a 95% confidence interval for k
- If we repeat this exact sampling and CI procedure many times, 95% of the intervals we construct will contain the true population value for k



Sampling Variance vs Confidence Intervals

Sampling variance quantifies the instability of a sample statistic

- How much the statistic bounces around between samples

A confidence interval estimates the plausible location of an population statistic

- We haven't yet said how to find a confidence interval

Next Time

- Bootstrapped confidence intervals